

Matplotlib Assignment

1. In order to assess the performance of students of a class in the annual examination, the class teacher stored marks of the students in all the 5 subjects in a CSV “Marks.csv” file as shown in Table. Plot the data using boxplot and perform a comparative analysis of performance in each subject.

Name	English	Maths	Hindi	Science	Social_Studies
Rishika Batra	95	95	90	94	95
Waseem Ali	95	76	79	77	89
Kulpreet Singh	78	81	75	76	88
Annie Mathews	88	63	67	77	80
Shiksha	95	55	51	59	80
Naveen Gupta	82	55	63	56	74
Taleem Ahmed	73	49	54	60	77
Pragati Nigam	80	50	51	54	76
Usman Abbas	92	43	51	48	69

2. Prayatna sells designer bags and wallets. During the sales season, he offered discounts ranging from 10% to 50% over a period of 5 weeks. He recorded his sales for each discount rate in an array as follows:

Discounts (%) = [10, 20, 30, 40, 50]

Sales (in Rs.) = [40000, 45000, 48000, 50000, 100000].

Draw a scatter plot to show the relationship between the discount offered and the sales made.

3. Analyze and visualize automobile fuel efficiency Auto MPG dataset using `pandas` for data manipulation and `matplotlib` for plotting.

Dataset URL: ``http://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data``

Column Names: `['mpg', 'cylinders', 'displacement', 'horsepower', 'weight', 'acceleration', 'model_year', 'origin', 'car_name']`

a. Data Loading and Cleaning

- Load the dataset using `pandas`.
- Assign proper column names.

- Handle missing values (hint: some rows have '?' in the `horsepower` column).
- Convert the `horsepower` column to numeric type.

b. Scatter Plot: Horsepower vs MPG

- Create a scatter plot showing the relationship between “horsepower” and “mpg”.
- Label the axes and title the plot appropriately.
- Use grid lines for better readability.

c. Histogram: Distribution of MPG

- Plot a histogram to show the frequency distribution of the `mpg` (miles per gallon).
- Use appropriate number of bins (e.g., 15–20).
- Label the axes and add a title.

d. Box Plot: (MPG by Cylinders)

- Create a box plot showing the distribution of “mpg” grouped by the number of “cylinders”.
- Each box should represent one cylinder group (e.g., 4, 6, 8).
- Add axis labels and a title.